

ANAND GOPALAKRISHNAN

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OBJECTIVE

I am a Postdoctoral Researcher in AI and Cognitive Science at Harvard University with Samuel Gershman and Yilun Du. I completed my PhD in Computer Science at IDSIA with Jürgen Schmidhuber. My research asks what architectural and learning principles enable neural networks to discover modular structure and abstractions inspired by human perception and cognition, and generalize efficiently beyond their training experience. I have worked across compositional representation learning, self-supervised learning, reinforcement learning, and sequence modeling, with the goal of building more capable, efficient, and robust AI systems.

EDUCATION

PhD in Informatics , Università della Svizzera Italiana (USI Lugano)	2019-2025
MS in Electrical Engineering , Pennsylvania State University	2017-2019
BTech in Electrical Engineering , National Institute of Technology Karnataka (NITK)	2012-2016

SELECTED PUBLICATIONS

- **A Gopalakrishnan**, R Csordás, J Schmidhuber, MC Mozer, "Decoupling The ‘What’ and ‘Where’ With Polar Coordinate Positional Embedding", *International Conference on Machine Learning (ICML)*, 2026. **Spotlight**
- E Littwin, V Thilak, **A Gopalakrishnan**, "Enhancing JEPAs with Spatial Conditioning: Robust and Efficient Representation Learning", *Workshop on Self-Supervised Learning, NeurIPS*, 2024.
- **A Gopalakrishnan**, A Stanić, J Schmidhuber, MC Mozer, "Recurrent Complex-Weighted Autoencoders for Unsupervised Object Discovery", *Neural Information Processing Systems (NeurIPS)*, 2024.
- K Irie, **A Gopalakrishnan**, J Schmidhuber, "Exploring the Promise and Limits of Real-time Recurrent Learning", *International Conference on Learning Representations (ICLR)*, 2024.
- A Stanić*, **A Gopalakrishnan***, K Irie, J Schmidhuber, "Contrastive Training of Complex-Valued Autoencoders for Object Discovery", *Neural Information Processing Systems (NeurIPS)*, 2023.
- J Gha, V Herrmann, B Grewe, J Schmidhuber, **A Gopalakrishnan**, "Unsupervised Musical Object Discovery from Audio", *Machine Learning for Audio Workshop, NeurIPS*, 2023.
- **A Gopalakrishnan**, K Irie, J Schmidhuber, S van Steenkiste, "Unsupervised Learning of Temporal Abstractions using Slot-based Transformers", *Neural Computation*, 35(4), 593-626, 2023.
- M Zhuge, H Liu, F Faccio, DR Ashley, R Csordás, **A Gopalakrishnan**, A Hamdi, H Hammoud, V Herrmann, K Irie, L Kirsch, B Li, G Li, S Liu, J Mai, P Piekos, A Ramesh, I Schlag, W Shi, A Stanić, W Wang, Y Wang, M Xu, DP Fan, B Ghanem, J Schmidhuber, "Mindstorms in Natural Language-based Societies of Mind", *Computational Visual Media*, 2023. **Best paper award at R0-FoMo NeurIPS Workshop 2023**
- **A Gopalakrishnan**, S van Steenkiste, J Schmidhuber, "Unsupervised Object Keypoint Learning using Local Spatial Predictability", *International Conference on Learning Representations (ICLR)*, 2021. **Spotlight**
- **A Gopalakrishnan**, A Mali, D Kifer, C.L Giles and A.G Ororbia, "A Neural Temporal Model for Human Motion Prediction" *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.

ACADEMIC POSITIONS

Postdoctoral Researcher in AI and Cognitive Science Harvard University	October 2025 - present Cambridge, MA
Scalable architectures that incorporate synchronization and oscillatory dynamics. Improving Transformers ability to perform state-tracking and serve as an episodic memory.	

EXPERIENCE

Research Intern Apple Designed architecture and pretraining improvements to Joint Embedding Predictive Architectures (JEPAs).	June 2024 - September 2024 Cupertino, CA
Applied Scientist Intern Amazon AWS AI Lab Investigated object-centric world models for interventional and counterfactual predictions.	June 2022 - November 2022 Tübingen, Germany
Research Assistant Indian Institute of Science (IISc Bangalore) Contributed to Indic-TTS project. A Text-to-Speech system for Indian Languages.	June 2016 - April 2017 Bangalore, India

AWARDS & HONORS

- Awarded compute grant of 150,000 GPU hours from Swiss National Supercomputing Centre for the project *Scaling Synchrony-based Models for Visual Binding* 2024.
- Selected for the Brains, Minds and Machines summer school 2023 by MIT Center for Brains, Minds and Machines.
- Awarded compute grant of 200,000 GPU hours from Swiss National Supercomputing Centre for the project *Learning Structured World Models for Visual Perception and Reasoning* 2020.
- Awarded NeurIPS D&I Travel award 2019.
- Awarded National Talent Search Examination Scholarship 2008. A competitive national-level scholarship given to 1000 high-school students in India.

SKILLS

Technical Skills	Deep learning, Self-supervised learning, Reinforcement learning, Transformers, World models, LLMs, VLMs, Multimodal perception
Programming Languages	Python, LaTeX, MATLAB, Bash
Frameworks and Tools	PyTorch, TensorFlow, Keras, Numpy, Scipy, Scikit-learn, OpenCV, Pandas, Git, Markdown, Inkscape
Miscellaneous	Scientific Writing, Public Speaking, Analytical Skills, Organization, Visual Art

PROFESSIONAL ACTIVITIES

Reviewing • Conferences: ICLR 2022 NeurIPS 2022, 2023 ICML 2023, 2026 (Top 25% Reviewer Award) • Journals: Neural Computation • Workshops: Elements of Reasoning: Objects, Structure and Causality at ICLR 2022 Object Representations for Learning and Reasoning at NeurIPS 2020 Meta-Learning Workshop at NeurIPS 2019
Teaching • Course: Graduate-level Machine Learning at USI Lugano. Designed lectures, tutorial sessions and assignments for unsupervised learning and probabilistic learning modules.

SUPERVISION

Joonsu Gha (Master's thesis) Title: Learning Structured Music Representations from Audio Data Now: PhD in Neuroscience & Machine Learning at ETH Zurich	August 2022 - August 2023
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